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A known mass of the substance is taken along with fuming HNO_3 and $AgNO_3$ is taken in a clean carius tube. The open end of the Carius tube is sealed and placed in a iron tube for 5 hours in the range at 530-540 k Then the tube is allowed to cool and a small hole is made in the tube to allow gases produced to escape. The tube is broken and the ppt is filtered, washed, dried and weighed. From the mass of AgX obtained, percentage of halogen in the organic compound is calculated.

Weight of the organic compound: w g

Weight of AgCl precipitate = a g

143.5 g of AgCl contains 35.5 g of Cl

$\therefore$ a g of AgCl contains $\frac{35.5}{143.5} \times a$

W g Organic compound gives a g AgCl

Percentage of Cl in w g $= \left(\frac{35.5}{143.5} \times \frac{a}{w} \times 100\right)\%$ organic compound

Let Weight of silver Bromide be 'b'g

188g of AgBr contains 80 g of Br

$\therefore$ b g of AgBr contains $\frac{80}{188} \times \frac{b}{w}$ of Br

W g Organic compound gives b g AgBr

Percentage of Br in w g $= \left(\frac{80}{188} \times \frac{b}{w} \times 100\right)\%$ organic compound

Let Weight of silver Iodide be 'c'g

235g of AgI contains 127 g of I

$\therefore$ C g of AgI contains $\frac{127}{235} \times \frac{c}{w}$ of I

W g Organic compound gives c g AgI

Percentage of I in w g $= \left(\frac{127}{235} \times \frac{c}{w} \times 100\right)\%$ organic compound

EXAMPLE : 0.284 g of an organic substance gave 0.287 g AgCl in a carius method for the estimation of halogen. Find the Percentage of Cl in the compound.

Weight of the organic substance = 0.284 g

Weight of AgCl is = 0.287 g

143.5 g of AgCl contains 35.5 g of chlorine

0.287 g of AgCl contains $\frac{35.5}{143.5} \times \frac{0.287}{0.284}$

% of chlorine is $\frac{35.5}{143.5} \times \frac{0.287}{0.284} \times 100 = 24.98$

Evaluate Yourself

7) 0.185 g of an organic compound when treated with Conc. HNO_3 and silver nitrate gave 0.320 g of silver bromide. Calculate the % of bromine in the compound. (Ag =108, Br = 80) Ans: 73.6

8) 0.40 g of an iodo-substituted organic compound gave 0.235 g of AgI by carius method. Calculate the percentage of iodine in the compound. (Ag = 108 I = 127) (Ans = 31.75)

Estimation of phosphorus:

Carius method: A known mass of the organic compound (w) containing phosphorous is heated with fuming HNO_3 in a sealed tube where C is converted into CO_2 and H to H_2O. phosphorous present in organic compound is oxidized to phosphoric acid which is precipitated, as ammonium phosphomolybdate by heating with Conc. HNO_3 and then adding ammonium molybdate.